TBvax 3.4.3 model extensions
==========

Summary

This report briefly describes the extensions of the TBvax / EndTB model developed so far for KNCV (see contract).

These model extensions include:

1. the development and transmission of drug resistant variants of Mtb
2. the application of diagnostic tests and follow-up treatment
3. TB preventive treatment

In addition, this report documents basic testing of the new features and suggest further follow-up tests. Note that major parts of this document are rather technical as descriptions are given on how to define the various features in the model specification.

1. Drug resistant TB

The model now support both the development of drug resistant Mtb strains (as a result of treatment) and the transmission of these strains.

Definition of drug resistant Mtb strains in the XML of EndTB

Drug resistant Mtb strains are defined in the model specification, i.e. the XML input file.

In the `<TB.infectivity>` XML element multiple `<infectivity.matrix>` elements can be specified, each for a specific Mtb strain which should be specified with the attribute `Mtb-strain`:

```
<TB.infectivity>
  <age.groups>
    <age.group name="A0"/>
    <age.group name="A15"/>
  </age.groups>
  <TB.parameter name="infs" value="0.83"/>
  <TB.parameter name="infc" value="1"/>
  <TB.parameter name="DRinf" value="1"/>
  <infectivity.matrix  Mtb-strain="drugsens">
    <infectivity stage="Ds" value="infs"/>
    <infectivity stage="Dc" value="infc"/>
  </infectivity.matrix>
  <infectivity.matrix  Mtb-strain="drugresist">
    <infectivity stage="Dsdr" value="DRinf*infs"/>
    <infectivity stage="Dcdr" value="DRinf*infc"/>
  </infectivity.matrix>
</TB.infectivity>
```

In the example above Ds and Dc are the subclininal and clinical stages of drug-sensitive TB whereas Dsdr and Dcdr are the corresponding stages of drug-resistant TB. The infectivity of the ‘drugresist’ strain in the subclinical (Ds) and clinical (Dc) stages can be adjusted by the multiplier DRinf.

The transmission of drug-sensitive and drug-resistant TB is specified in the `<transmission.matrix>` elements in `<TB.transmission>`:

```
<TB.transmission>
```

```
    <age.groups>
      <age.group name="A0"/>
      <age.group name="A15"/>
    </age.groups>
    <TB.parameter name="DRmult" value="2"/>
    <TB.parameter name="pT" value="0.001522282"/>
    <transmission.matrix Mtb-strain="drugsens">
      <transition from="Un" to="Lf" rate="pT"/>
      <transition from="Ls" to="Lf" rate="pT"/>
    </transmission.matrix>
    <transmission.matrix Mtb-strain="drugresist">
      <transition from="Un" to="Lfdr" rate="DRmult*pT"/>
      <transition from="Lsdr" to="Lfdr" rate="DRmult*pT"/>
    </transmission.matrix>
  </TB.transmission>
```

The transmission probability pT of the drug resistant Mtb strain is multiplied with DRmult.

The development of drug resistance can be defined to be due to treatment:

```
    <treatment.matrix name="init">
      <transition from="Dc" to="T" rate="0.5*j4*eta/max(0.01,(1-pE))"/>
      <transition from="Dc" to="T_DR" rate="0.5*j4*eta/max(0.01,(1-pE))"/>
      <multiplier name="etamul" times="1960:2019" values="1/(1+exp(-sh*(x-midx)))"/>
    </treatment.matrix>
```

In the example above, treatment of clinical TB (Dc) leads to development of drug resistant (as T_DR is the ‘on treatment’ stage of drug resistant TB from which failed treatment will lead to a transition to Dc_DR.

A full example is given in
`./countries/ZAF/parameters/XMLinput_2910_m_DSDR_noHIV.xml`
which is used in the R script
`./code-demos/example-ZAF-DRTB-noHIV.R`

Note that in this example, HIV is set to 0 to keep the XML relatively simple.

Tests

Tested so far:

- development of DR due to treatment
- transmission of DR strain

2. Diagnostic tests and follow-up treatment

Diagnostic tests are defined in the XML inputfile in the new <test> elements within the <tests> section:

```
<tests>
<test name="CXR">
     <TB.stage.tested name="Un" fraction.positive="0.041"/>
      <TB.stage.tested name="L0" fraction.positive="0.042"/>
      <TB.stage.tested name="Lf" fraction.positive="0.043"/>
      <TB.stage.tested name="Ls" fraction.positive="0.044"/>
      <TB.stage.tested name="Ds" fraction.positive="0.25"/>
      <TB.stage.tested name="Dc" fraction.positive="0.975"/>
</test>
<test name="CULT">
      <TB.stage.tested name="Un" fraction.positive="0.005"/>
      <TB.stage.tested name="L0" fraction.positive="0.005"/>
      <TB.stage.tested name="Lf" fraction.positive="0.005"/>
      <TB.stage.tested name="Ls" fraction.positive="0.005"/>
      <TB.stage.tested name="Ds" fraction.positive="1"/>
      <TB.stage.tested name="Dc" fraction.positive="1"/>
</test>
<test name="SYMPTOMS">
      <TB.stage.tested name="Un" fraction.positive="0.1"/>
      <TB.stage.tested name="L0" fraction.positive="0.1"/>
      <TB.stage.tested name="Lf" fraction.positive="0.1"/>
      <TB.stage.tested name="Ls" fraction.positive="0.1"/>
      <TB.stage.tested name="Ds" fraction.positive="0.1"/>
      <TB.stage.tested name="Dc" fraction.positive="0.9"/>
</test>
<test name="IGRA">
      <TB.stage.tested name="Un" fraction.positive="0.02"/>
      <TB.stage.tested name="L0" fraction.positive="0.02"/>
      <TB.stage.tested name="Lf" fraction.positive="0.83"/>
      <TB.stage.tested name="Ls" fraction.positive="0.83"/>
      <TB.stage.tested name="Ds" fraction.positive="0.83"/>
      <TB.stage.tested name="Dc" fraction.positive="0.81"/>
</test>
<test name="TST">
  <TB.stage.tested name="Un" fraction.positive="0.05"/>
  <TB.stage.tested  name="L0" fraction.positive="0.89"/>
  <TB.stage.tested  name="Lf" fraction.positive="0.89"/>
  <TB.stage.tested  name="Ls" fraction.positive="0.89"/>
  <TB.stage.tested  name="Ds" fraction.positive="0.89"/>
  <TB.stage.tested  name="Dc" fraction.positive="0.75"/></test>
<compound.test name="TBpos"   expression="(IGRA + SYMPTOMS * (1-IGRA)) * CXR * CULT"/>
<compound.test name="LTBIpos" expression="IGRA * (1 - CXR * CULT)"/>
</tests>
```

In the example above the TB states are (as defined elsewhere in the XML):
Un: uninfected
L0: latent after clearing TB infection
Lf: latent fast
Ls: latent slow
Ds: subclinical TB
Dc: clinical TB

Each test defines the fraction positively tested for each of the TB states.
In addition to defining simple tests, it is possible to defined composed tests:

```
<compound.test name="TBpos"   expression="(IGRA + SYMPTOMS * (1-IGRA)) * CXR * CULT"/>
<compound.test name="LTBIpos" expression="IGRA * (1 - CXR * CULT)"/>
```

The above compound tests `TBpos` and `LTBIpos` are defined by the combination of simple tests.
The evaluation of the expressions would work as follows:

TB stage	IGRA	SYMPTOMS	CXR	CULT	TBpos=(IGRA+SYMPTOMS*(1-IGRA))*CXR*CULT	LTBIpos=IGRA*(1-CXR*CULT)
Un	0.02	0.1	0.041	0.005	0.00002419	0.0199959
L0	0.02	0.1	0.042	0.005	0.00002478	0.0199958
Lf	0.83	0.1	0.043	0.005	0.000182105	0.82982155
Ls	0.83	0.1	0.044	0.005	0.00018634	0.8298174
Ds	0.83	0.1	0.25	1	0.21175	0.6225
Dc	0.81	0.9	0.975	1	0.956475	0.02025

To use the result of a positive test in treatment, the option `after.test` can be used in the definition of the treatment matrix:

```
<treatment.matrix name="init" after.test="TBpos">
  <transition from="Dc" to="T" rate="(1+aHIVtreatinit*th1)*j4*eta/max(0.01,(1-pE))"/>
  <multiplier name="etamul" times="1960:2020" values="1/(1+exp(-sh*(x-midx)))"/>
</treatment.matrix>
```

In the example above only those tested positive (i.e. about 96% of Dc) would be treated.
Not included (yet) in the example is the effect of treating Ds (of which 21% would test positive).
This would require an additional transition e.g.

```
  <transition from="Ds" to="T" rate="(1+aHIVtreatinit*th1)*j4*eta/max(0.01,(1-pE))"/>
```

Working examples can be found in:
`./countries/ZAF/XMLinput_2910_m_diag_test_treat_noHIV.xml`
`./countries/ZAF/XMLinput_2910_m_TPT_noHIV_after_LTBI_test.xml`
which is called from:
`./code-demos/example-ZAF-diag-test-treat-noHIV.R`

Tests
Tested so far:

- correct fraction tested positive and negative
- correct evaluation of expression for compound test
- correct fraction treated after (compound) test

3. TB preventive treatment (TPT)

TPT depends on the number of diagnosed TB cases (i.e. the number in Dc multiplied with a rate (e.g. the number with starting treatment) and the average household size.

This is specified in the XML as follows:

```
<treatment.matrix name="init">
  <transition from="Dc" to="T" rate="eta"/>
  <multiplier name="etamul" times="1960:2019" values="1/(1+exp(-sh*(x-midx)))"/>
</treatment.matrix>
<treatment.matrix name="TPT" trigger="TB:Dc">
  <transition from="Ls" to="TPT" rate="(hhsize-1)*eta"/>
  <transition from="Lf" to="TPT" rate="(hhsize-1)*eta"/>
  <multiplier name="etamul" times="1960:2019" values="1/(1+exp(-sh*(x-midx)))"/>
</treatment.matrix>
```

The treatment matrix for TPT specifies the attribute `trigger="TB:Dc"` which means that the number treated with preventive therapy will depend on the number in Dc multiplied with `eta` (the treatment initiation rate, see treatment matrix `"init"`) and `(hhsize-1)`.
The flow from Ls and Lf to TPT will also depend on the fraction of Ls and Lf in the total population.

For example, with `eta=0.40` and average household size `hhsize=3.5` the rate of entering TPT equals `(3.5-1)*0.4 = 1.0`. With Dc = 75 the number of people entering TPT would be 75 / yr if everyone were in the Lf and Ls stages. Multiplying with the fraction of people in Ls and Lf (about 1 in 6) means that about 12 people per year start TPT.

As TBvax does not simulate households, the effect of TPT on TB incidence will underestimate the true effect of TPT as in a household with a household member in Dc, it is more likely that other household members are in Lf or Ls.

Working examples:
`./countries/ZAF/parameters/XMLinput_2910_m_TPT_noHIV_hhsize_noSES.xml`
i.e. TPT as a result from diagnosing TB in the Dc stage (for simplicity SES removed/disabled)

`./countries/ZAF/parameters/XMLinput_2910_m_TPT_noHIV_hhsize.xml`
i.e. TPT as a result from diagnosing TB in the Dc stage with SES

`./countries/ZAF/parameters/XMLinput_2910_m_TPT_HIV_hhsize_noSES_duetoHIV.xml`
i.e. TPT due to PLHIV (for simplicity SES removed/disabled)

`./countries/ZAF/parameters/XMLinput_2910_m_TPT_noHIV_after_LTBI_test.xml`
i.e. TPT due to diagnosing TB in the Dc stage and TPT only given after positive LTBI test

These examples are called from:
`./code-demos/example-ZAF-TPT.R`

Tests

tested so far:
- correct fractions of Ls and Lf treated with TPT
- correct combination with compound test (e.g. LTBI)
- correct fractions of Ls and Lf treated with TPT due to HIV